

# Ayan Baruah

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## EDUCATION

|                                                                                                                       |                                     |
|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>Indian Institute of Technology (ISM) Dhanbad</b><br><i>Bachelor of Technology in Civil Engineering (GPA: 8.10)</i> | Expected May 2028<br>Dhanbad, India |
| <b>South Point School, Guwahati</b><br><i>Class XII (CBSE) - (Percentage: 90.2%)</i>                                  | 2024<br>Guwahati, India             |
| <b>Don Bosco School, Guwahati</b><br><i>Class X (CBSE) - (Percentage: 97.2%)</i>                                      | 2022<br>Guwahati, India             |

## PROJECTS

**JobVector** — MERN, TypeScript, Gemini API | [Live Demo](#) | [GitHub](#) | **AI Impact Summit Hackathon 2026 — Honourable Mention**

- Architected a **bi-directional RAG pipeline** that intelligently recommends jobs to candidates and ranks applicants for recruiters using a multi-stage retrieval system and **LLM-based reranking**.
- Implemented **Hybrid Search with RRF** (Reciprocal Rank Fusion) in Pinecone, combining dense semantic vectors and sparse keywords to maximize matching accuracy.
- Engineered a multimodal resume parser by streaming raw PDFs directly to an LLM, extracting factual candidate details to feed the semantic vector pipeline.

**CodeArena** — Coding Platform with Custom Contest Rooms | MERN, JavaScript, Docker | [Live Demo](#) | [GitHub](#)

- Developed a full-stack coding contest platform using the MERN stack, featuring real-time **Socket.IO-based** contest rooms where users can **create custom contests**, compete with friends, and interact via live chat.
- Engineered a secure, **scalable Docker-based sandbox** for isolated code execution and asynchronous submission processing using **BullMQ and Redis**.
- Built Google Docs-style real-time collaborative code editing with **Yjs CRDTs**, Socket.IO, and the Monaco Editor, supporting live cursors, awareness, and conflict-free synchronization.

**Road SOS** — Next.js, React, TypeScript | [Live Demo](#) | [GitHub](#)

- Developed and deployed a cross-platform emergency web-app, significantly reducing accident response time by **automating SOS triggers** and routing users to the nearest hospitals and police stations.
- Engineered a Capacitor Motion G-force collision detector instantly **transmitting the victim's live location to emergency contacts**, with a 10s cancel buffer and offline native SMS fallback.

## TECHNICAL SKILLS

**Languages:** C, C++, Python, JavaScript, TypeScript, HTML5  
**Frameworks:** React.js, Next.js, Express.js, Node.js, Tailwind CSS  
**Databases:** MongoDB, Pinecone, MySQL, PostgreSQL, Redis  
**Tools:** Git/GitHub, Postman, Firebase, Docker, Kubernetes, BullMQ, VS Code, Cloud AI  
**Course Works:** Data Structures & Algorithms, Object Oriented Programming, DBMS, Operating Systems  
**Interests:** Full Stack Development, Competitive Programming, Generative AI, Machine Learning

## ACHIEVEMENTS

- Codeforces Rating: 1408 (Specialist) [Codeforces Profile](#)
- Solved 300+ Data Structures and Algorithms problems on LeetCode, Codeforces.
- Strong foundation in data structures, algorithms, and performance-optimized problem solving.

## EXTRA-CURRICULAR ACTIVITIES

**Sports:** Badminton (State-level) | Represented IIT (ISM) Dhanbad at **Inter IIT Sports Meet 2024 & 2025**